

Name:-Aditi Yedge.**Roll No:- 68****Batch:-A2****Subject:- DEVOPS**

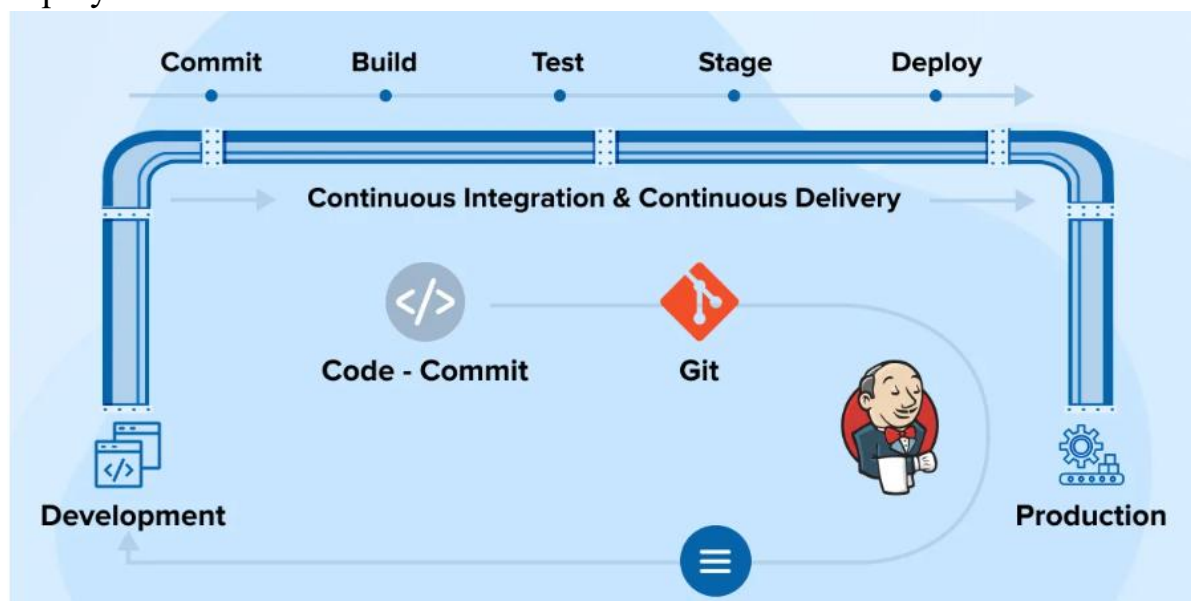
Assignment 3

Aim: Demonstrate Continuous Integration and development using Jenkins.

What is Jenkins Pipeline?

Jenkins Pipeline (or simply "Pipeline" with a capital "P") is a suite of plugins which supports implementing and integrating *continuous delivery pipelines* into Jenkins.

A *continuous delivery (CD) pipeline* is an automated expression of your process for getting software from version control right through to your users and customers. Every change to your software (committed in source control) goes through a complex process on its way to being released. This process involves building the software in a reliable and repeatable manner, as well as progressing the built software (called a "build") through multiple stages of testing and deployment.



Step 1. Download the Jenkins WAR File.

Click on the [Jenkins download](#) for Windows.

The image shows a web browser window displaying the Jenkins download page for Windows. The page title is "Thank you for downloading Windows Stable installer". It includes a link "Download hasn't started? Click this link". Below this, there are sections for "Changing boot configuration", "Starting/stopping the service", and "Inheriting your existing Jenkins installation". At the bottom, there is a "See Also" section with links to "Running Jenkins behind Internet Information Server (IIS)", "Running Jenkins behind nginx", and "Running Jenkins behind Apache".

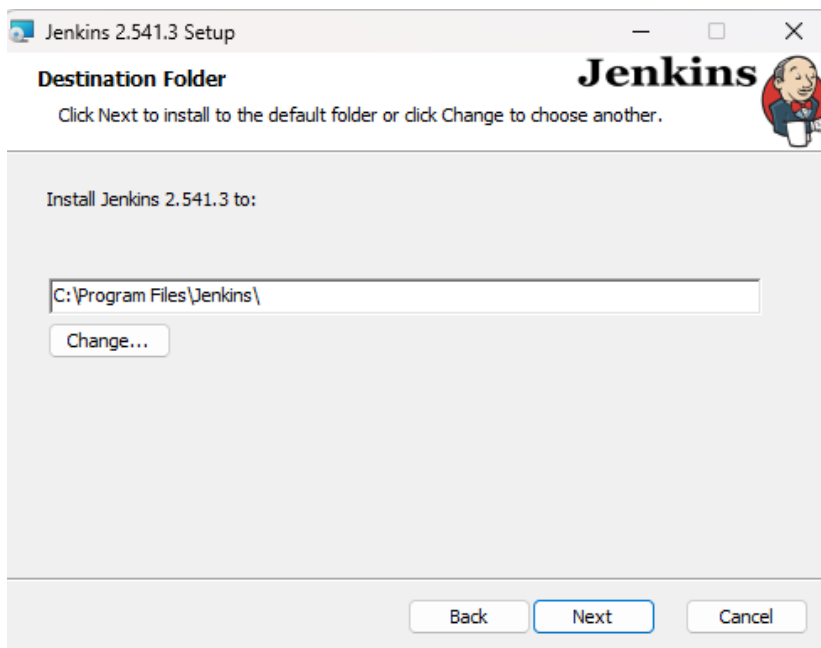
Below the browser window, a Windows File Explorer window is open, showing the "Downloads" folder. The file "jenkins" is visible in the "Last week" section.

Step2: Setup wizard

When you open the Windows Installer, an **Installation Setup Wizard** appears. Click **Next** on the Setup Wizard to start the installation.

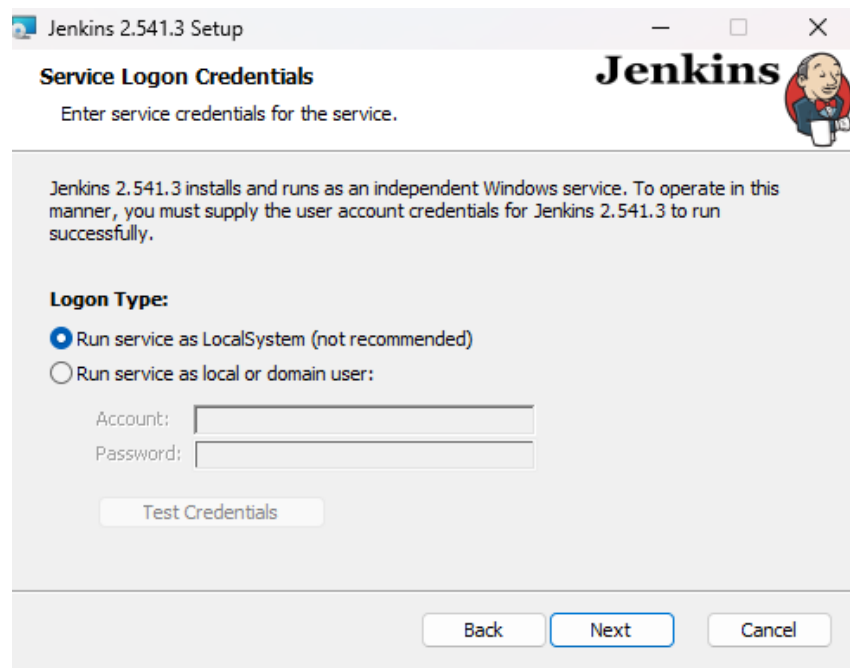
**Step 2: Select destination folder**

Select the destination folder to store your Jenkins Installation and click **Next** to continue.



Step 3: Service logon credentials

When Installing Jenkins, it is recommended to install and run Jenkins as an independent Windows service using a **local or domain user**, as it is much safer than running Jenkins using **LocalSystem(Windows equivalent of root)** which will grant Jenkins full access to your machine and services. To run Jenkins service using a **local or domain user**, specify the domain user name and password with which you want to run Jenkins, click on **Test Credentials** to test your domain credentials and click on **Next**.



Step 4: Port selection

Specify the port on which Jenkins will run. Use the **Test Port** button to validate whether the specified port is free on your machine.

Consequently, if the port is free, it will show a green tick mark as shown below. Then, click on **Next**.

Jenkins 2.541.3 Setup

Port Selection

Choose a port for the service.

Please choose a port.

Port Number (1-65535):

8082

Test Port

⚠ Click 'Test Port' button to proceed

It is recommended that you accept the selected default port.

Back Next Cancel

Jenkins 2.541.3 Setup

Port Selection

Choose a port for the service.

Please choose a port.

Port Number (1-65535):

8082

Test Port

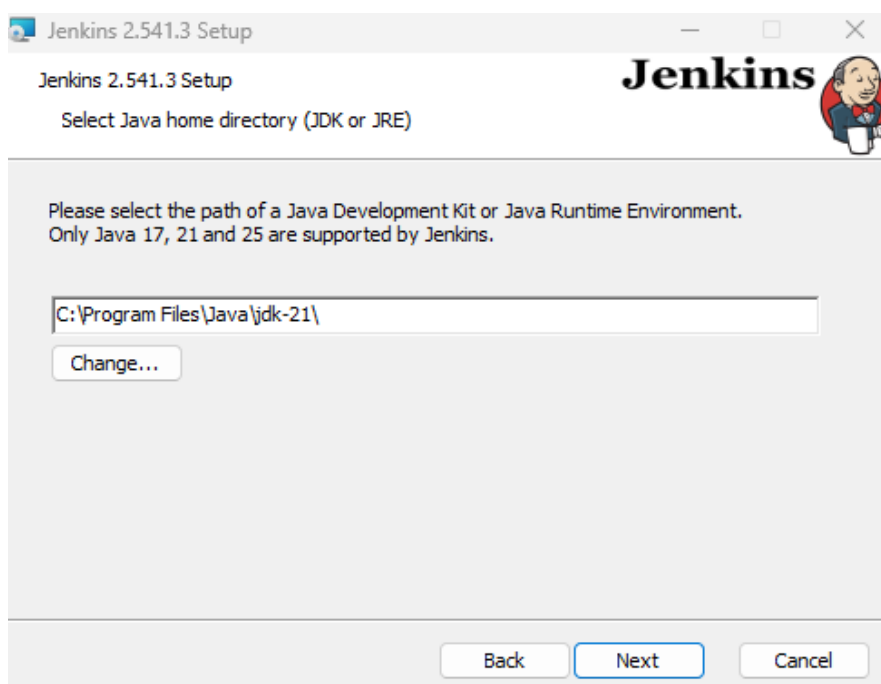
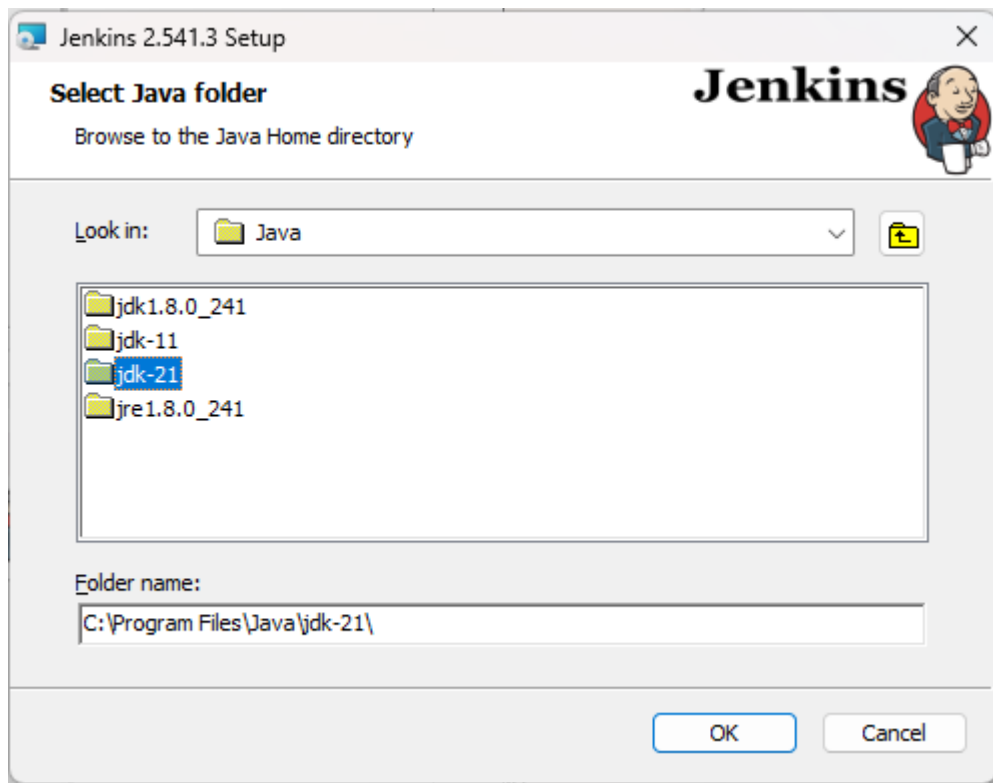
✓

It is recommended that you accept the selected default port.

Back Next Cancel

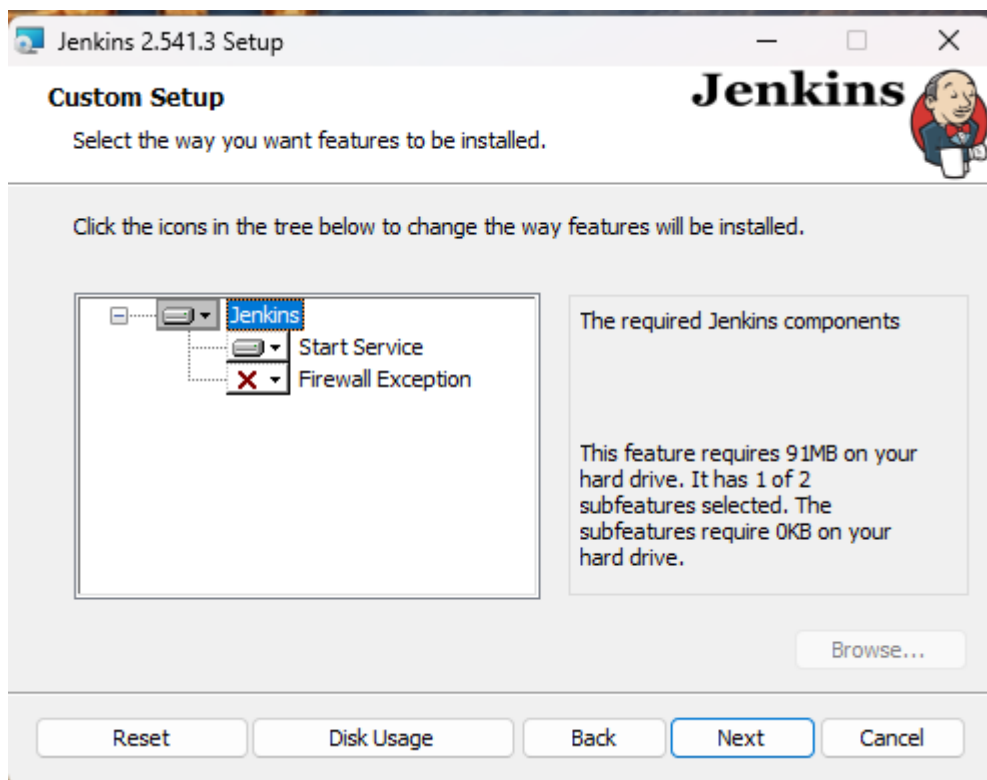
Step 5: Select Java home directory

The installation process checks for Java on your machine and prefills the dialog with the Java home directory. If the needed Java version is not installed on your machine, you will be prompted to install it. Once your Java home directory has been selected, click on **Next** to continue.

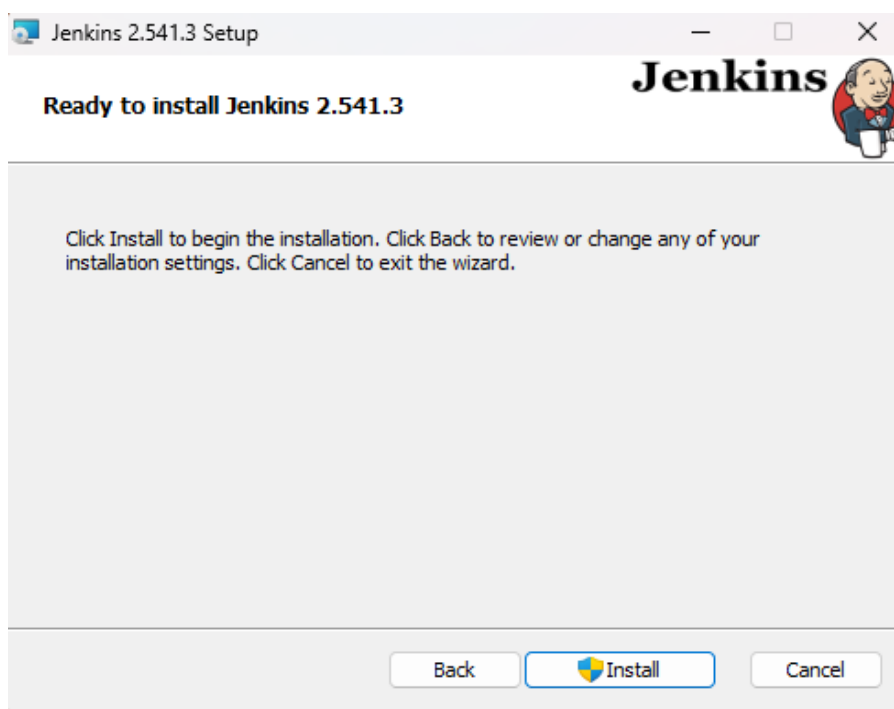


Step 6: Custom setup

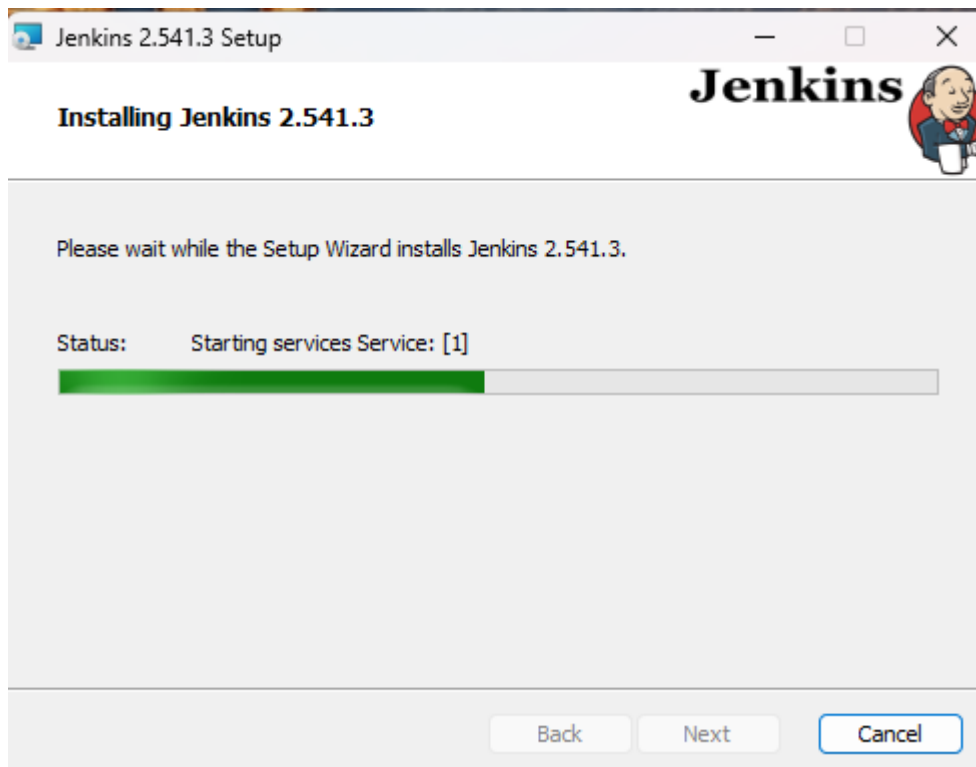
Select other services that need to be installed with Jenkins and click on **Next**.

**Step 7: Install Jenkins**

Click on the **Install** button to start the installation of Jenkins.

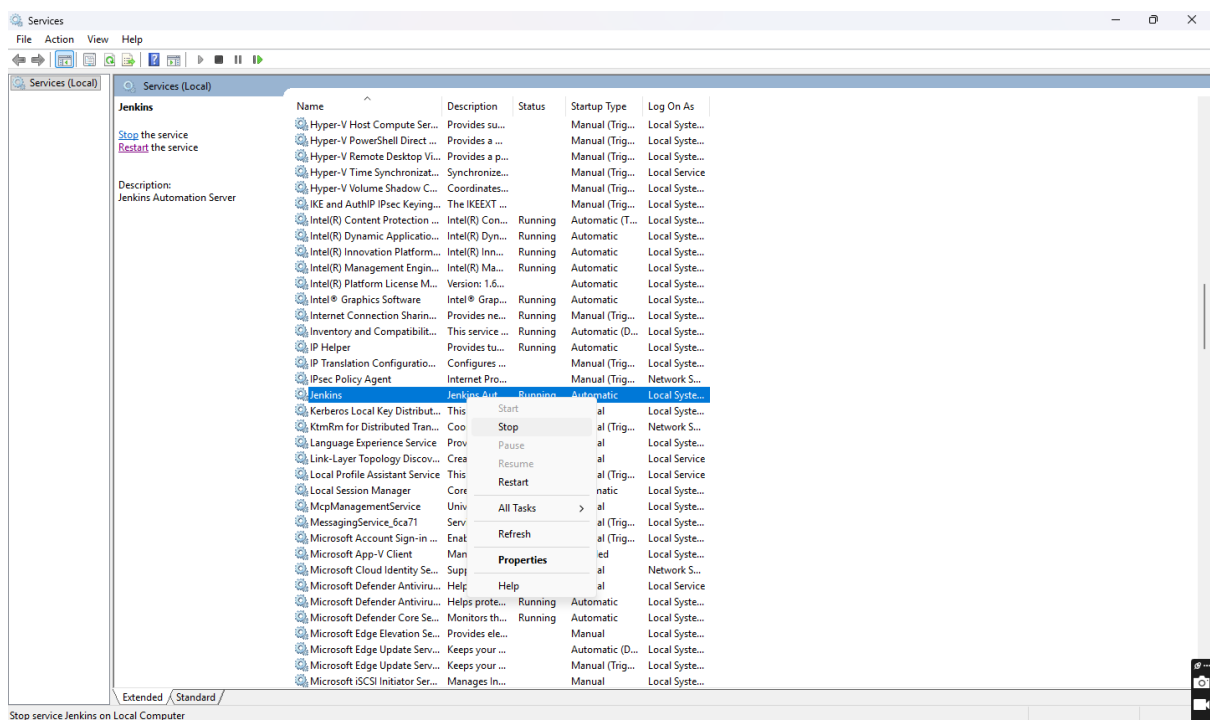


Additionally, clicking on the **Install** button will show the progress bar of installation, as shown below:



Step 8: Finish Jenkins installation

Once the installation completes, click **Finish** to complete the setup





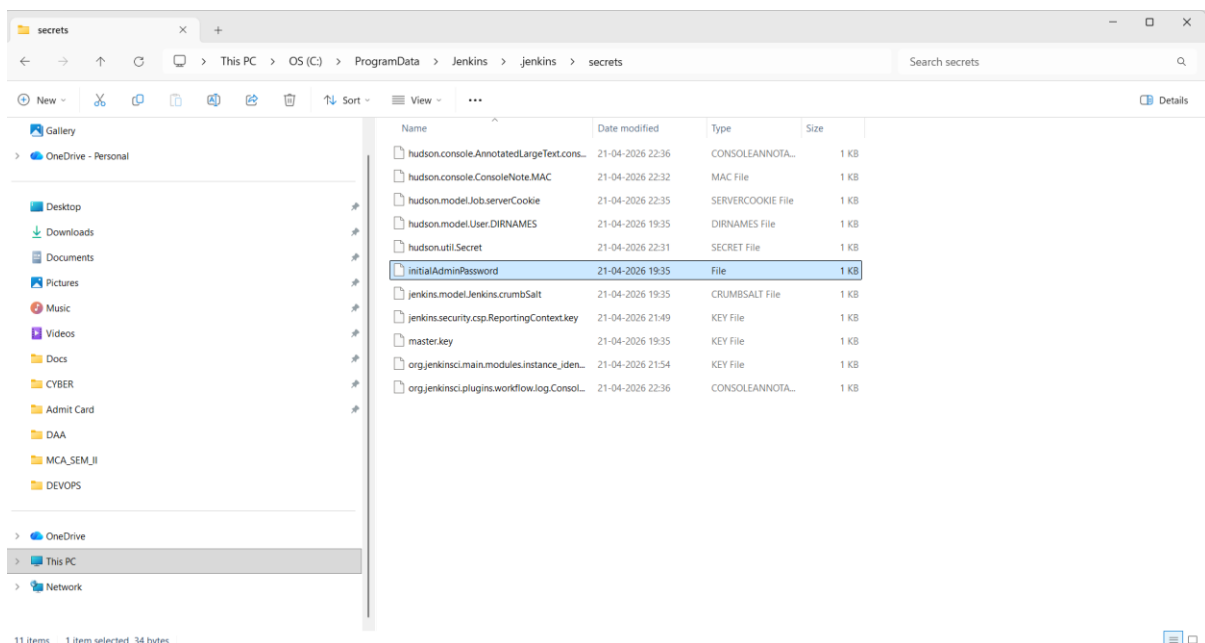
Step 9:- Sign in to jenkins

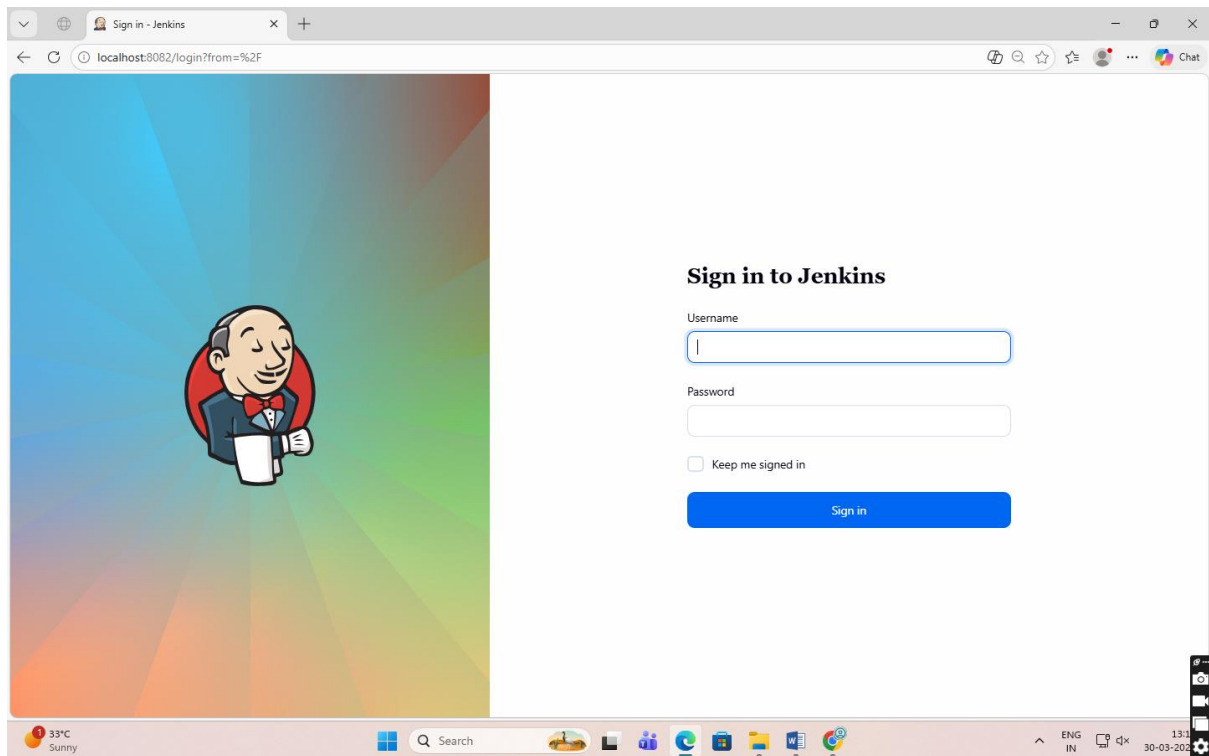
Use Username:-Admin

Password :- Follow Path

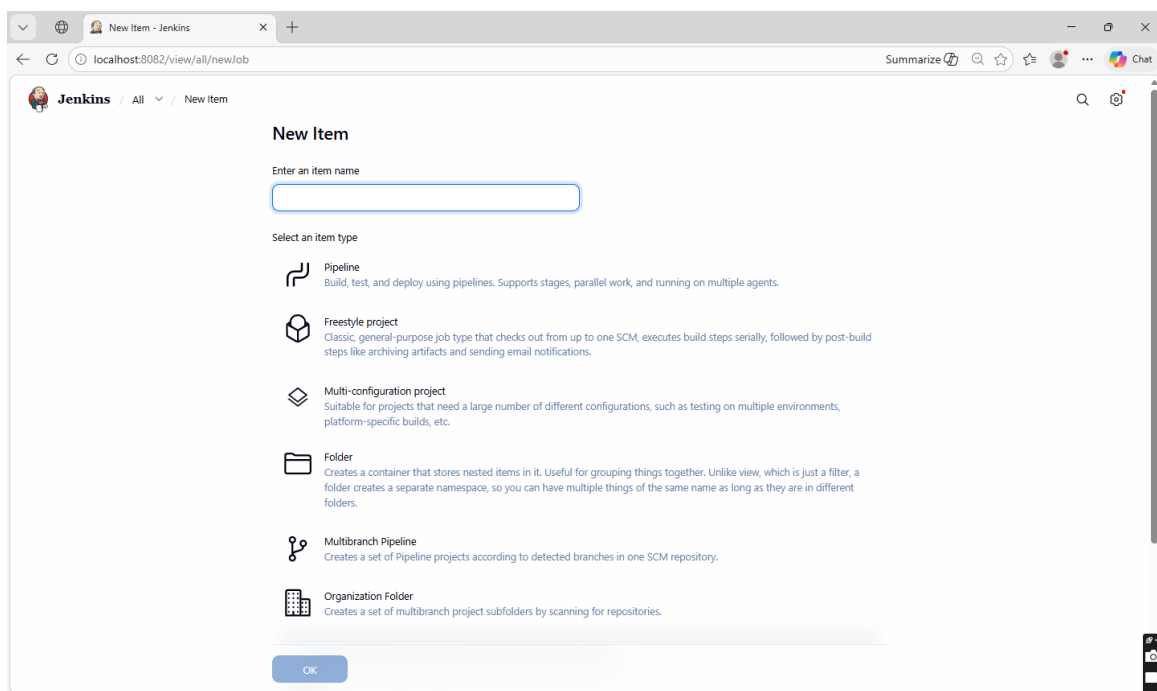
Program Data(S drive Hidden folder) -> Jenkins->.jenkins -> Secrets -> initialadminpassword

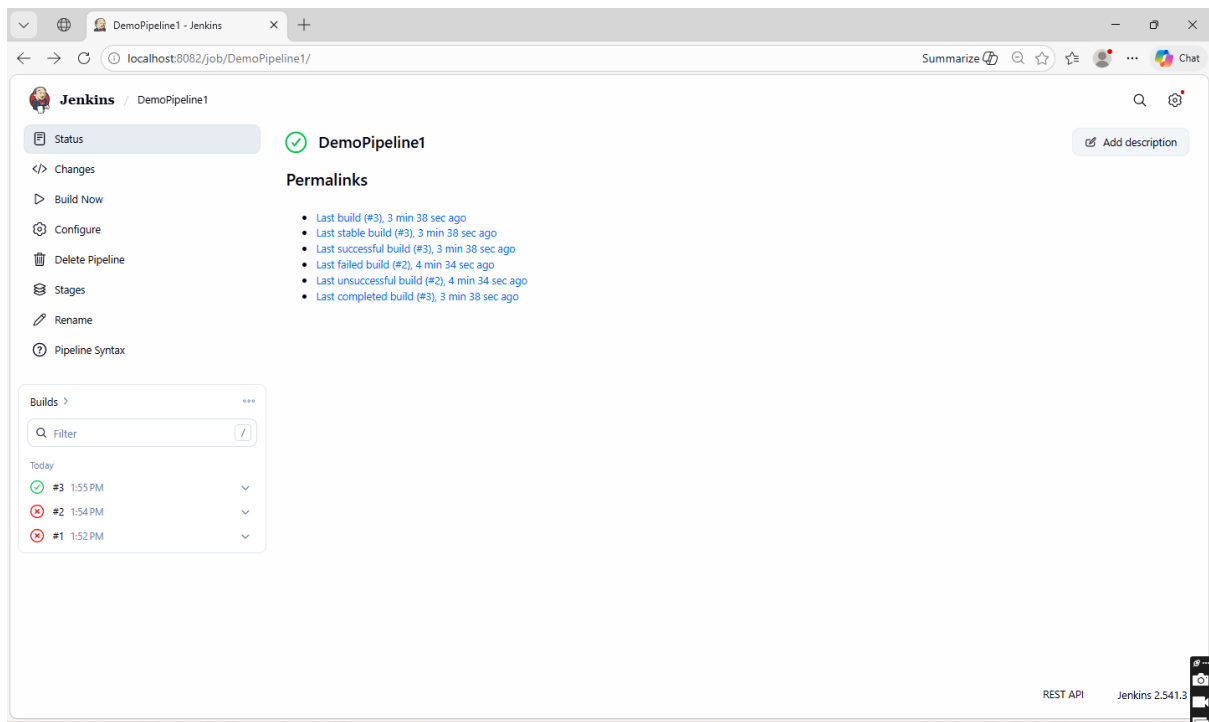
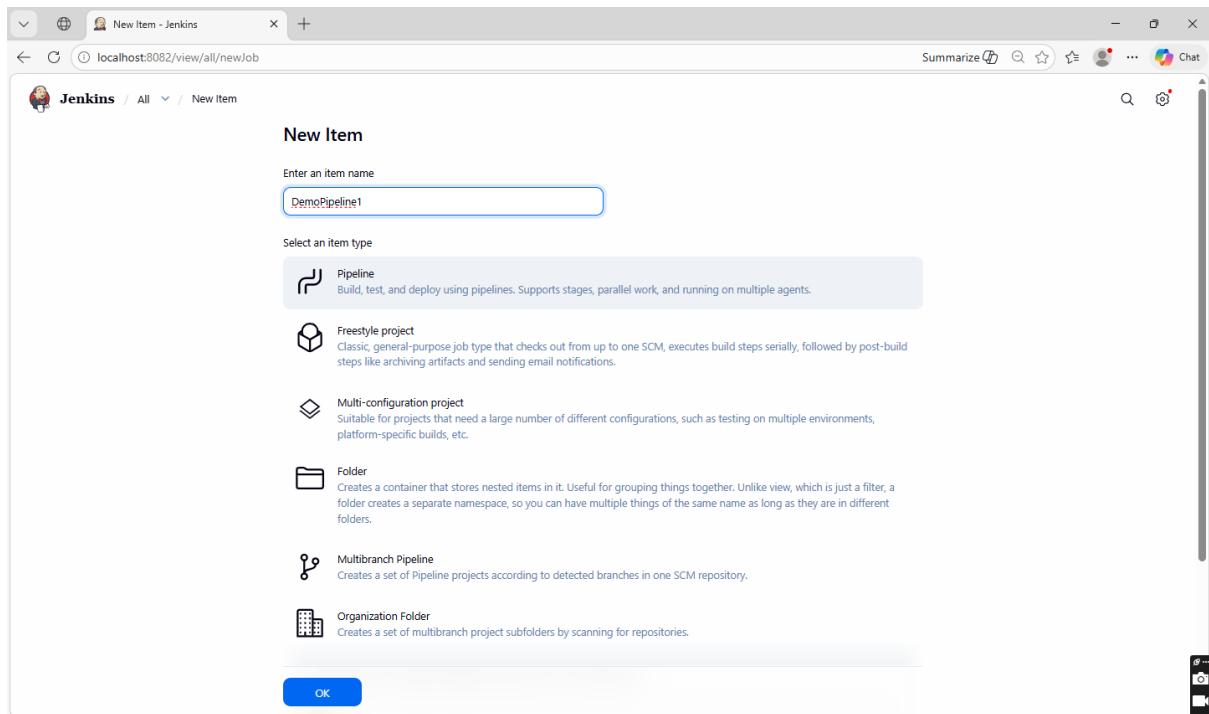
C:\ProgramData\Jenkins\.jenkins\secrets





Step1:- New Item → Demo Script →Write Script→Build Now

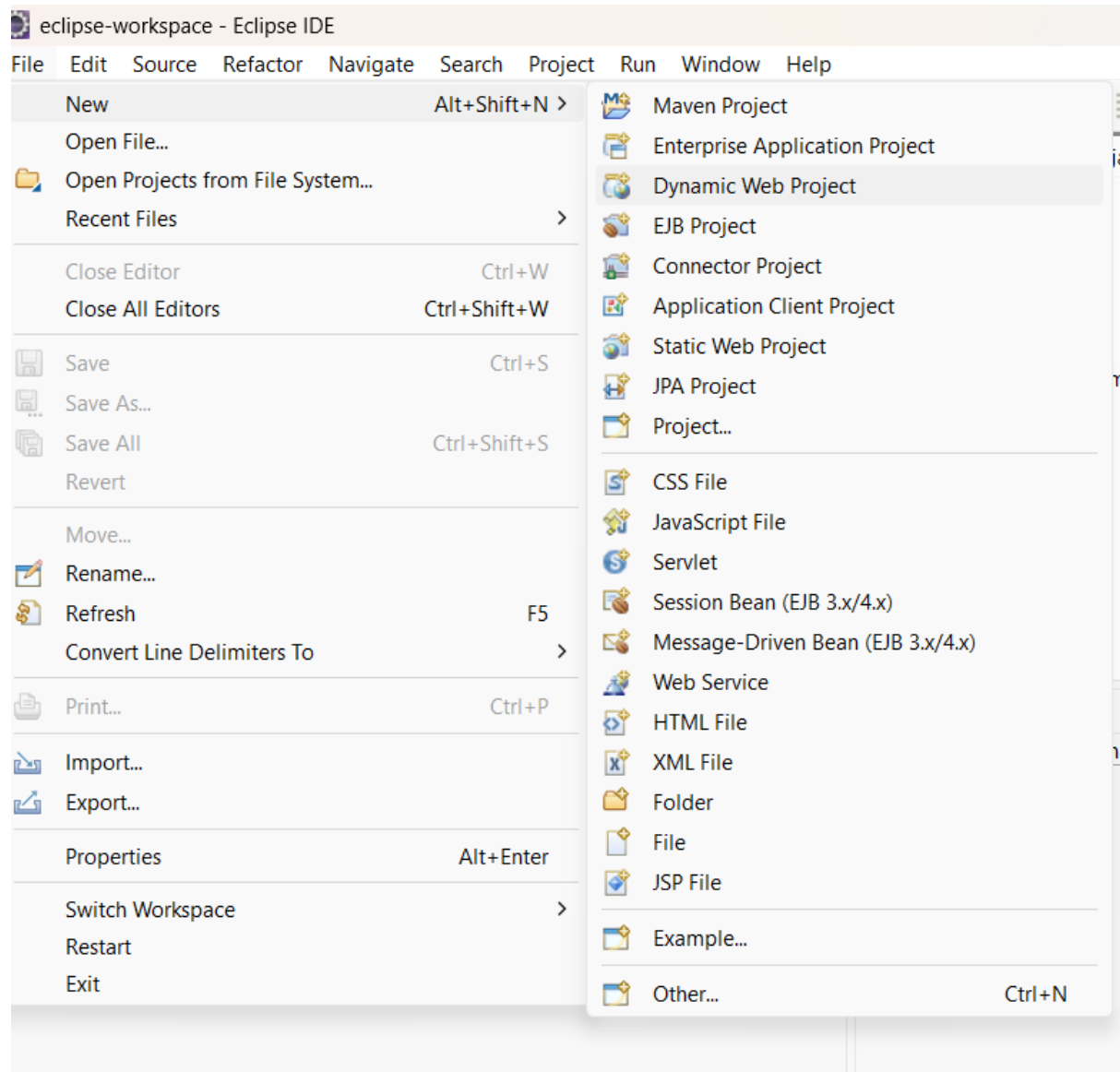


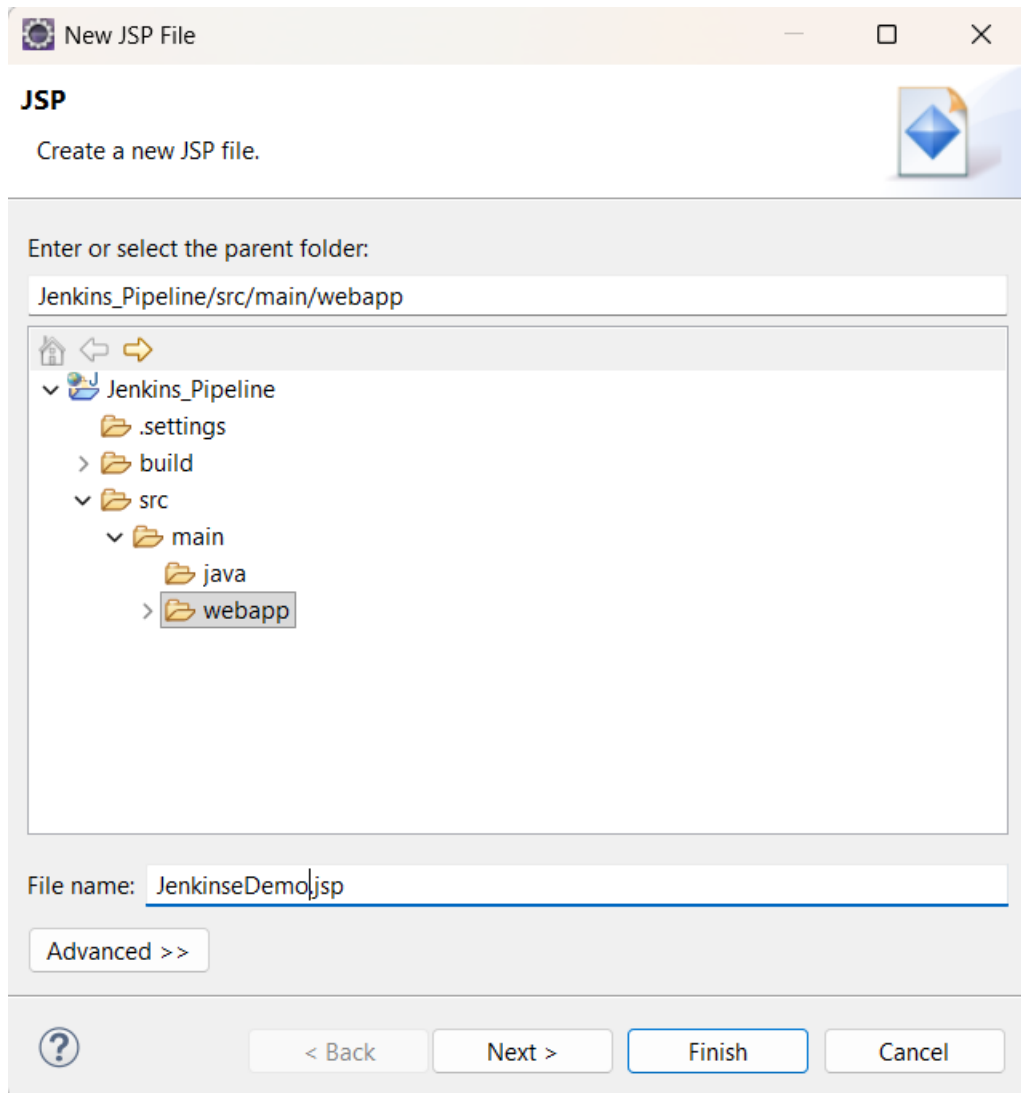


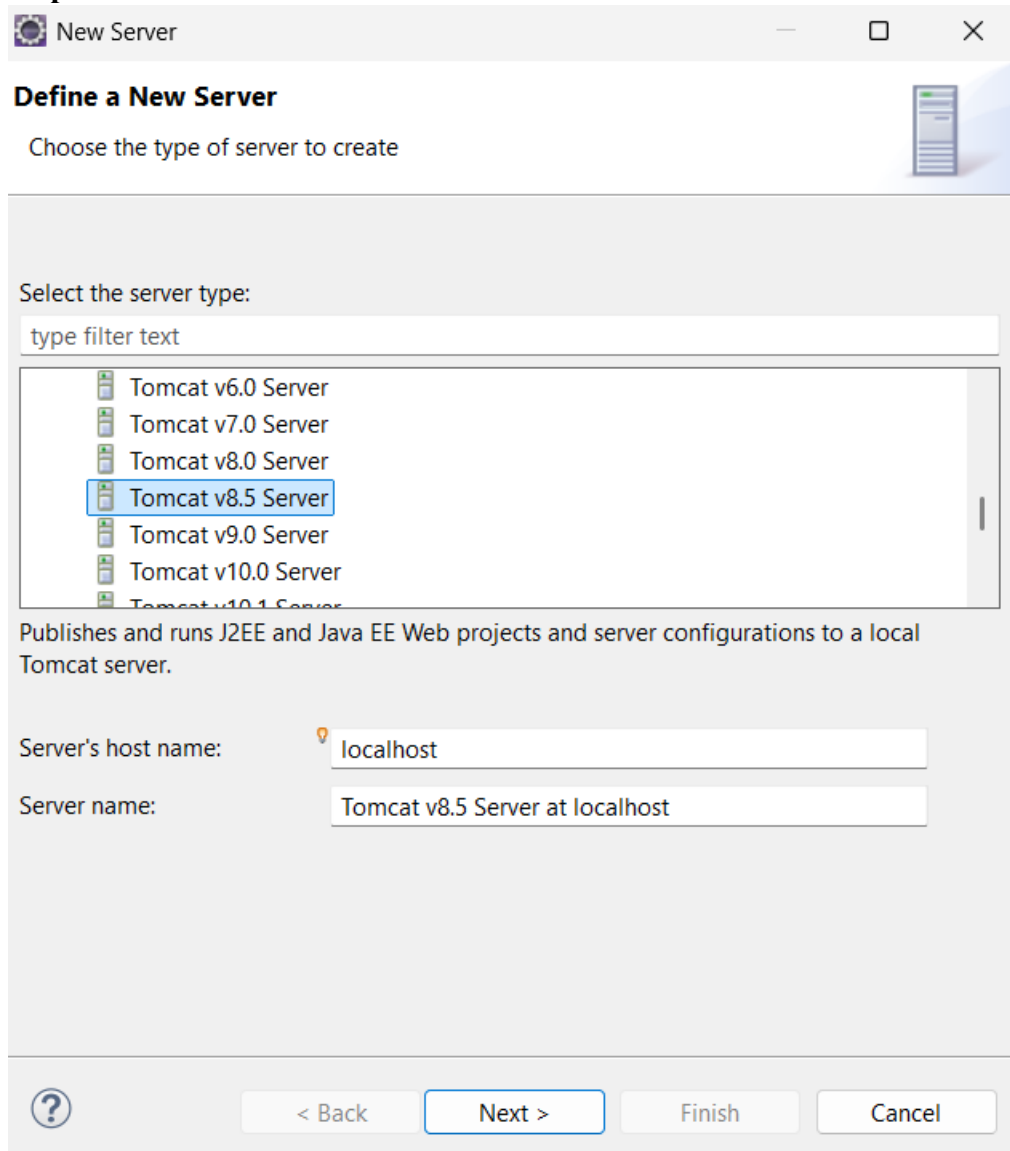
The screenshot shows the Jenkins job status page for 'DemoPipeline1 #3'. The browser address bar indicates the URL is 'localhost:8082/job/DemoPipeline1/3/'. The page title is 'Jenkins / DemoPipeline1 / #3'. The status is 'Success' (green checkmark) for build '#3 (Mar 30, 2026, 1:55:30 PM)'. The left sidebar contains links: Status, Changes, Console Output, Edit Build Information, Delete build '#3', Timings, Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Previous Build. The main content area shows 'Started by anonymous user' and 'This run spent: 13 ms waiting; 1.6 sec build duration; 1.6 sec total from scheduled to completion.' Below this, it says 'No changes.' There are buttons for 'Add description' and 'Keep this build forever'. The bottom right corner shows 'REST API' and 'Jenkins 2.541.3'.

The screenshot shows the Jenkins console output for 'DemoPipeline1 #3'. The browser address bar indicates the URL is 'localhost:8082/job/DemoPipeline1/3/console'. The page title is 'Jenkins / DemoPipeline1 / #3'. The console output is displayed in a text area with buttons for 'Download', 'Copy', and 'View as plain text'. The output text is as follows:

```
Started by user unknown or anonymous
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\jenkins\workspace\DemoPipeline1
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the project..
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running tests..
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application..
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline completed successfully
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
```

#Jenkin Pipeline Using Web Application**Step1:- Create Dynamic WEB Project in Eclipse.**



Step 3:- Select Server.

New Server

Define a New Server

Choose the type of server to create

Select the server type:

type filter text

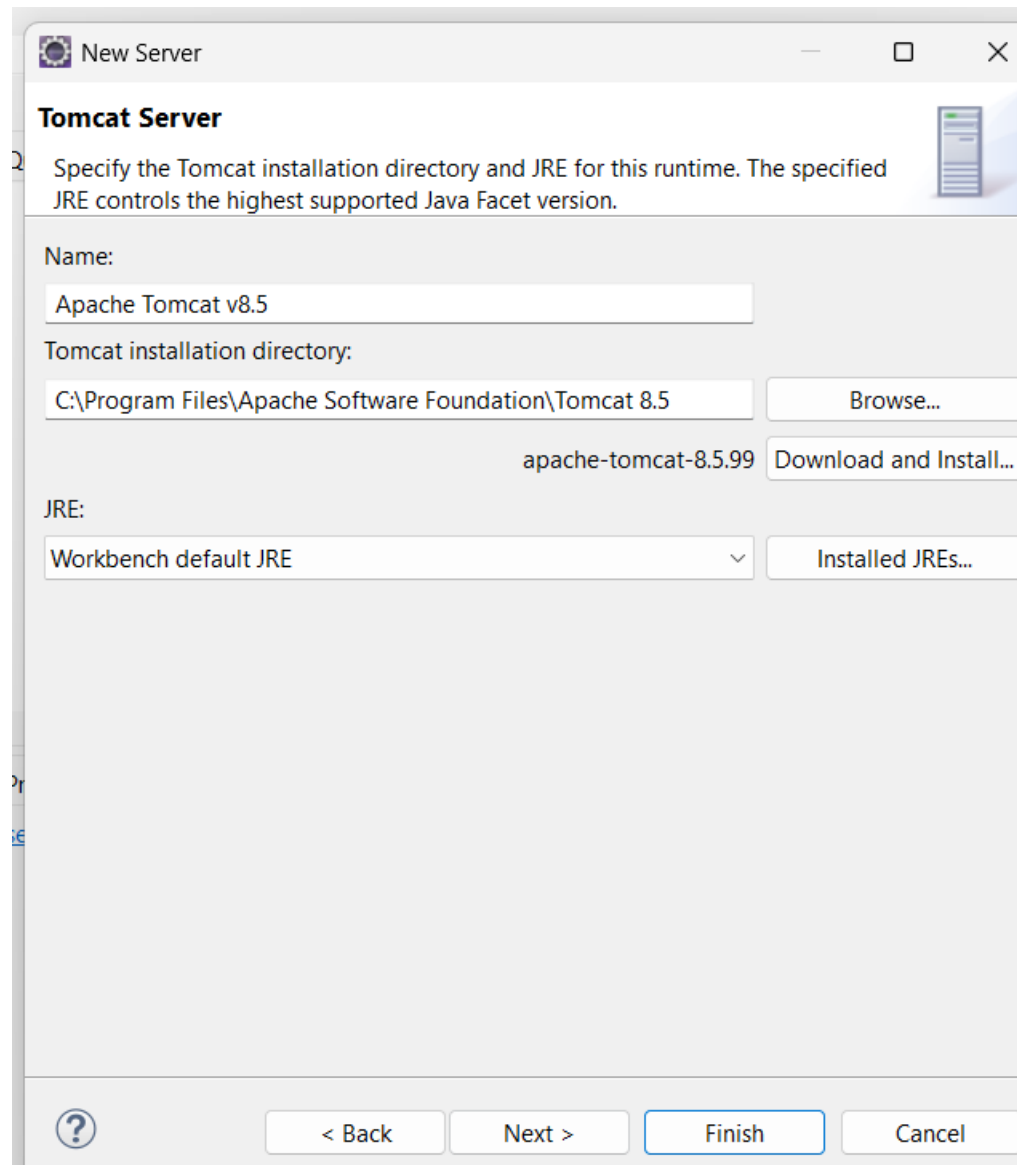
- Tomcat v6.0 Server
- Tomcat v7.0 Server
- Tomcat v8.0 Server
- Tomcat v8.5 Server**
- Tomcat v9.0 Server
- Tomcat v10.0 Server
- Tomcat v10.1 Server

Publishes and runs J2EE and Java EE Web projects and server configurations to a local Tomcat server.

Server's host name: localhost

Server name: Tomcat v8.5 Server at localhost

? < Back Next > Finish Cancel



New Server

Tomcat Server

Specify the Tomcat installation directory and JRE for this runtime. The specified JRE controls the highest supported Java Facet version.

Name:
Apache Tomcat v8.5

Tomcat installation directory:
C:\Program Files\Apache Software Foundation\Tomcat 8.5 Browse...

apache-tomcat-8.5.99 Download and Install...

JRE:
Workbench default JRE Installed JREs...

? < Back Next > Finish Cancel

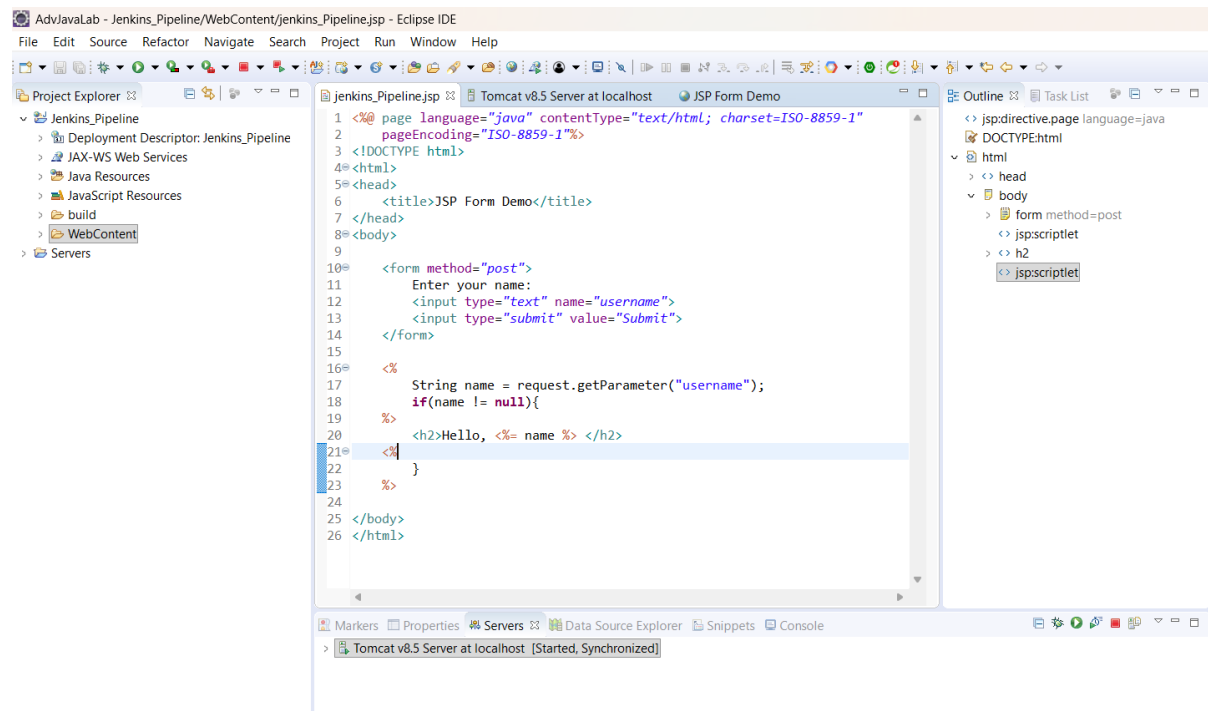
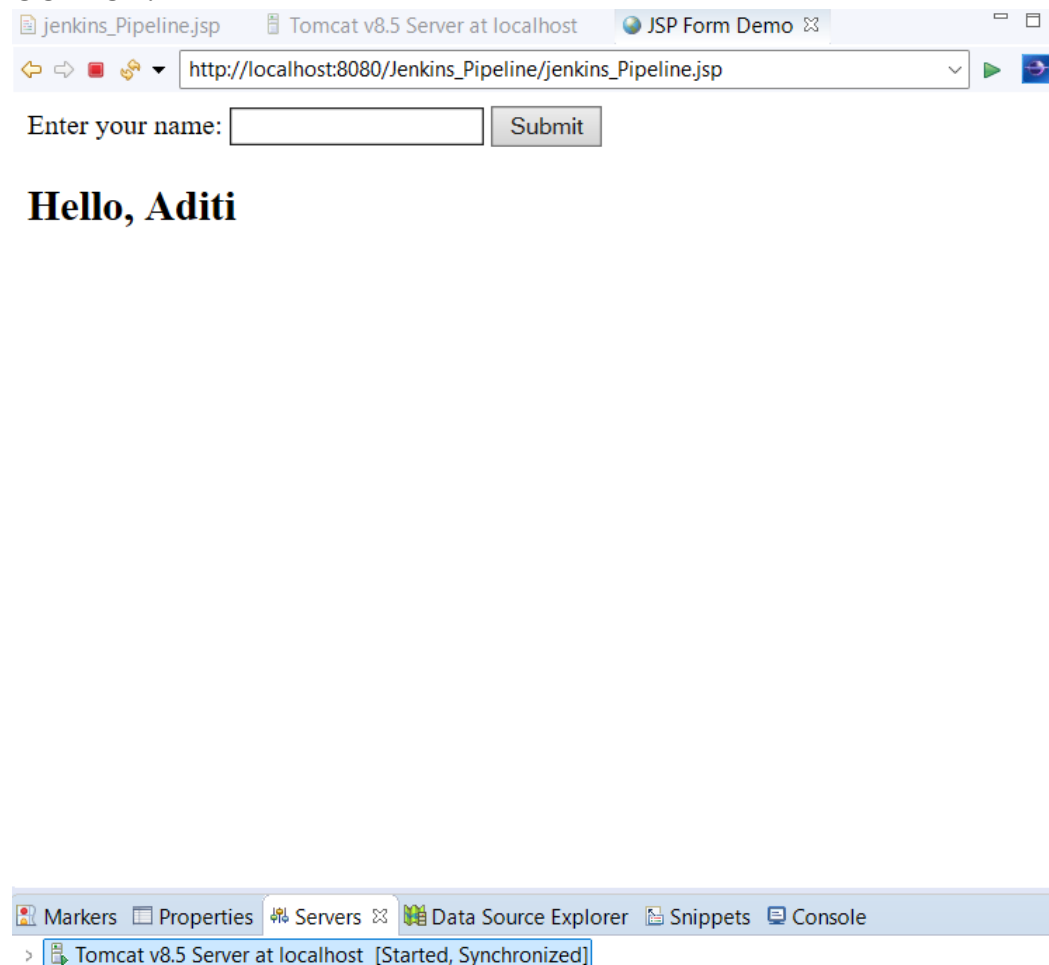
SourceCode:-

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<head>
    <title>JSP Form Demo</title>
</head>
<body>

    <form method="post">
        Enter your name:
        <input type="text" name="username">
        <input type="submit" value="Submit">
    </form>

    <%
        String name = request.getParameter("username");
        if(name != null){
    %>
        <h2>Hello, <%= name %> </h2>
    <%
        }
    %>

</body>
</html>
```

**OUTPUT:-**

Step 4:- Log in to GitHub Account.

1.Cretae A New Repository.

github.com/new

New repository

Search Type to search

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * yedgeaditi10 / Repository name * Jenkins_Pipeline
Jenkins_Pipeline is available.

Great repository names are short and memorable. How about [studious-memory](#)?

Description
0 / 350 characters

2 Configuration

Choose visibility * Public
Choose who can see and commit to this repository

Add README
READMEs can be used as longer descriptions. [About READMEs](#) Off

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#) No .gitignore

Add license
Licenses explain how others can use your code. [About licenses](#) No license

Create repository

© 2026 GitHub, Inc. Terms Privacy Security Status Community Docs Contact Manage cookies Do not share my personal information

Step 5:- Open Git Bash & follow the commands.

```
MINGW64:/c/Users/Adii/Desktop/DEVOPS/Jenkins_Pipeline

Adii@DESKTOP-C60SFL6 MINGW64 ~ (master)
$ cd "Desktop\DEVOPS\Jenkins_Pipeline"

Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git init
Reinitialized existing Git repository in C:/Users/Adii/Desktop/DEVOPS/Jenkins_Pipeline/.git/

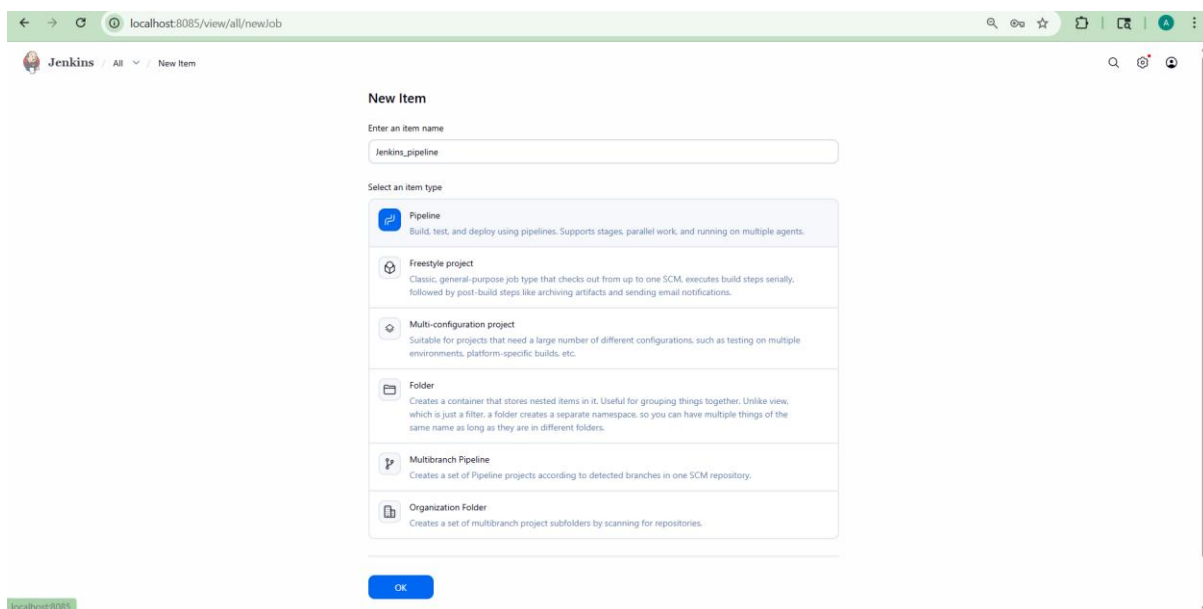
Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git add .

Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git remote add origin https://github.com/yedgeaditi10/Jenkins_Pipeline.git
error: remote origin already exists.

Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git branch -M main

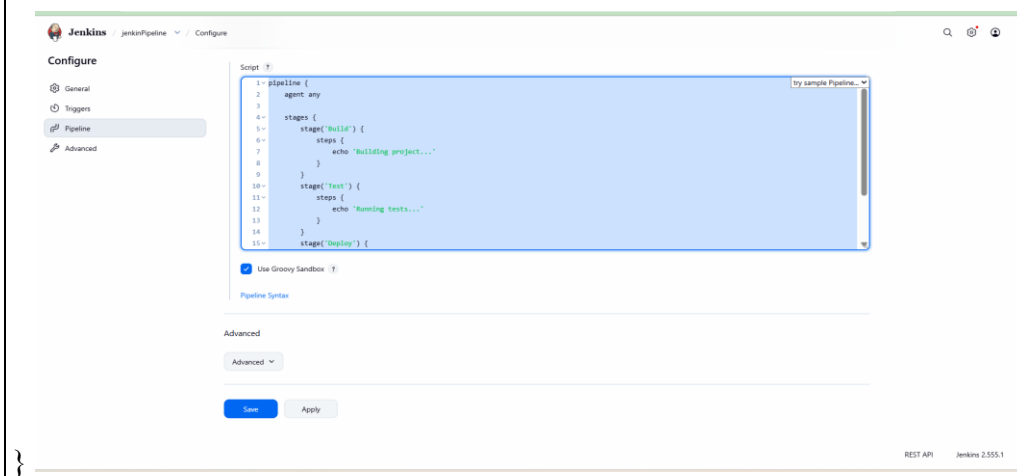
Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 8 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (16/16), 2.60 KiB | 296.00 KiB/s, done.
Total 16 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/yedgeaditi10/Jenkins_Pipeline.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Step 6:- Create a New Item And Select Pipeline



Write Script

```
pipeline {  
    agent any  
  
    stages {  
        stage('Build') {  
            steps {  
                echo 'Building project...'  
            }  
        }  
        stage('Test') {  
            steps {  
                echo 'Running tests...'  
            }  
        }  
        stage('Deploy') {  
            steps {  
                echo 'Deploying application...'  
            }  
        }  
    }  
}
```



OUTPUT:-



Jenkins / jenkinPipeline



Status



jenkinPipeline



Changes



Build Now



Configure



Delete Pipeline



Stages



Rename



Pipeline Syntax

Permalinks

- Last build (#1), 5 min 38 sec ago
- Last stable build (#1), 5 min 38 sec ago
- Last successful build (#1), 5 min 38 sec ago
- Last completed build (#1), 5 min 38 sec ago

Builds >



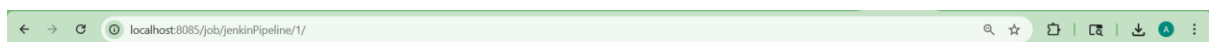
Filter



Today



#1 10:35 pm



Jenkins / jenkinPipeline / #1



Status



#1 (21-Apr-2026, 10:35:11pm)

Add description

Keep this build forever



Changes



Started by user admin



Console Output



This run spent:



Edit Build Information



- 52 ms waiting:
- 4.9 sec build duration:
- 5 sec total from scheduled to completion.



Delete build #1



Timings



Pipeline Overview



Restart from Stage



Replay



Pipeline Steps



Workspaces



No changes.

← → ↻

localhost:8085/job/jenkinsPipeline/1/console

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Jenkins

jenkinPipeline

#1

🔍 🔄 📄

Status

Changes

Console Output

Edit Build Information

Delete build #1

Timings

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Console

Download

Copy

View as plain text

Started by user admin

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in C:\ProgramData\jenkins\workspace\jenkinPipeline

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Build)

[Pipeline] echo

Building project...

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Test)

[Pipeline] echo

Running tests...

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Deploy)

[Pipeline] echo

Deploying application...

[Pipeline] }

[Pipeline] // stage

[Pipeline] }

[Pipeline] // node

[Pipeline] End of Pipeline

Finished: SUCCESS

Jenkins 2.555.1

Jenkins

jenkinPipeline

#1

🔍 🔄 📄

Started by admin

Started 4 min 27 sec ago

Queued 25 ms

Took 4.9 sec

🔄 Rerun

⚙️

Graph

Start

Build

Test

Deploy

End

🔍 Search

🔍

Build 0.22s

Test 0.1s

Deploy 0.12s

Deploy

0.12s

Started 4m 23s ago

Jenkins

🔍

⋮

Deploying application...

31ms

📄

0 Deploying application...

Jenkins 2.555.1

← → ↻

localhost:8085/job/jenkinsPipeline/1/flowGraphTable/

🔍 ☆ 🗑️ 📄 📄 📄 ⋮

Jenkins

jenkinPipeline

#1

🔍 🔄 📄

Status

Changes

Console Output

Edit Build Information

Delete build #1

Timings

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Step	Arguments	Status
Start of Pipeline - (4 sec in block)		🟢
node - (1.5 sec in block)		🟢
node block - (1 sec in block)		🟢
stage - (0.35 sec in block)	Build	🟢
stage block (Build) - (0.22 sec in block)		🟢
echo - (25 ms in self)	Building project...	🟢
stage - (0.26 sec in block)	Test	🟢
stage block (Test) - (0.1 sec in block)		🟢
echo - (25 ms in self)	Running tests...	🟢
stage - (0.21 sec in block)	Deploy	🟢
stage block (Deploy) - (0.12 sec in block)		🟢
echo - (31 ms in self)	Deploying application...	🟢

Jenkins 2.555.1